

Iliass Sijelmassi

Paris, France | +33 6 48 74 04 04 | iliass.sijelmassi-idrissi@polytechnique.edu | iliasssijelmassi.com | github.com/iliassSjm

SUMMARY

MSc Data Science & AI candidate (École Polytechnique & HEC Paris, GPA 3.8/4.0), Stanford visiting researcher applying ML to large-scale time-series prediction, and incoming Quantitative Researcher Intern at **WorldQuant**; seeking full-time **quantitative research** roles from Spring 2027.

EDUCATION

- École Polytechnique — MSc Data Science & AI (Year 1) GPA: 3.8/4.0 Sep 2024 – Jun 2025
 - Coursework: Probability & Statistics, Linear Algebra & Regression, Optimization, Deep Learning, Machine Learning.
- HEC Paris — MSc Data Science & AI (Year 2) Sep 2025 – Jun 2026
 - Quant-relevant coursework: Machine Learning for Financial Markets, Time Series, Causal Inference, Reinforcement Learning, Interpretability & Algorithmic Fairness, ML Ops & Cloud Engineering.
- INP-ENSEEIH, Toulouse — Dipl. Eng. in CS, Telecom & Applied Math Sep 2020 – Sep 2023
 - Coursework: Probability, Statistics, Optimization, Reinforcement Learning, Algorithms & Data Structures, High-Performance Computing.

EXPERIENCE

- Quantitative Researcher Intern | WorldQuant, Paris Oct 2026 – Mar 2027
 - Incoming 6-month internship in systematic **alpha signal research** and backtesting.
- Visiting Student Researcher | Stanford University, Stanford, CA Mar 2026 – Sep 2026
 - Building cardiac-event prediction models under Dr. Louise Sun on a **1.5 TB** longitudinal clinical dataset (ECG, echocardiogram, and arterial-pressure waveforms); **first-author paper in preparation**.
 - Designed a leakage-controlled, multi-horizon evaluation benchmarking **XGBoost / Random Forest** and deep models against clinical risk scores, with strict time-ordered splits, informative-missingness encoding (**MIA**), and **FFT**-based waveform features.
- Data Scientist Intern | Crédit Agricole CIB (CACIB), Paris Mar 2025 – Sep 2025
 - Modeled whether internal audit recommendations would be implemented on time across **200k+** historical records, reaching **F1 $\approx$ 0.75** and improving **ROC-AUC** over the heuristic baseline.
 - Designed a dual-model approach — **regularized Logistic Regression** for the coefficient-level interpretability auditors required, plus **XGBoost** for performance; drove feature selection with domain experts and validated robustness against **overfitting on out-of-time data** in **Python**, **Dataiku**, and **MLflow**.
- Software Engineering Intern | Infosys – Renault, Paris Mar 2023 – Oct 2023
 - Built backend modules for supply-chain platforms in **Java** and **PostgreSQL**.

QUANTITATIVE RESEARCH & ENGINEERING PROJECTS

- Crypto Perpetuals Alpha Strategy | ML for Financial Markets, HEC Paris 2026
 - Ranked **#1 of 13 teams (~50 students)** on out-of-sample Sharpe for a dollar-neutral long-short strategy across **411 cryptocurrency perpetual futures** (gross of transaction costs).
 - Engineered **379** candidate features — **316** retained after **Spearman IC** filtering and correlation pruning — into a **Ridge + XGBoost** ensemble under a rolling 12-month **walk-forward**, with PCA, scaler, and models refit per window and the IC filter fit once on train and frozen as an explicit anti-leakage choice.
- Leukemia Survival Prediction — QRT / ENS Data Challenge | Cox PH, Python 2025
 - Predicted overall survival on clinical + genomic data from 24 hospitals (**4,500+ patients**) using a **Cox Proportional Hazards** model; finished **top 12%**.
- Cognitive Alpha — Spatial Decision Analytics, FIFA World Cup 2022 | Personal Project, Python 2025–2026
 - Quantified, for each open-play pass, the gap between the player’s decision and the spatially optimal alternative: trained an **Expected Threat (xT)** model via **Markov-chain value iteration** on **126k possession actions**, with a NumPy-vectorised pitch-control engine and a **Streamlit** dashboard over **~5.4 GB** of tracking data.

TECHNICAL SKILLS & ACTIVITIES

- Programming: Python (primary), SQL (PostgreSQL), R, Bash; C++ / Java (working knowledge)
- Quantitative & ML: Probability & Statistics, Time-Series, Survival Analysis, Signal/Alpha Research
- Tools: PyTorch, XGBoost, Scikit-learn, NumPy, Pandas, MLflow, Docker, Git, Kafka, FastAPI, Dataiku, Linux
- Self-study & Certs: Stochastic calculus & options theory (Hull, Natenberg, Wilmott); Akuna Capital Options 101 & 202; Jane Street puzzles
- Languages: French (Native), English (Fluent); Dutch, Spanish, Portuguese & Arabic (B2)
- Activities: Built and sold a **200k-follower** X/Twitter account to **Polymarket**; private mathematics tutor (3+ years); Chess (1700 Elo)